How to accelerate the convergence of nonlinear solvers?

Rémy Vallot - Lightning talk

Application of Digital Twins to Large-Scale Complex Systems,
IMSI, Chicago, December 1 — 5, 2025

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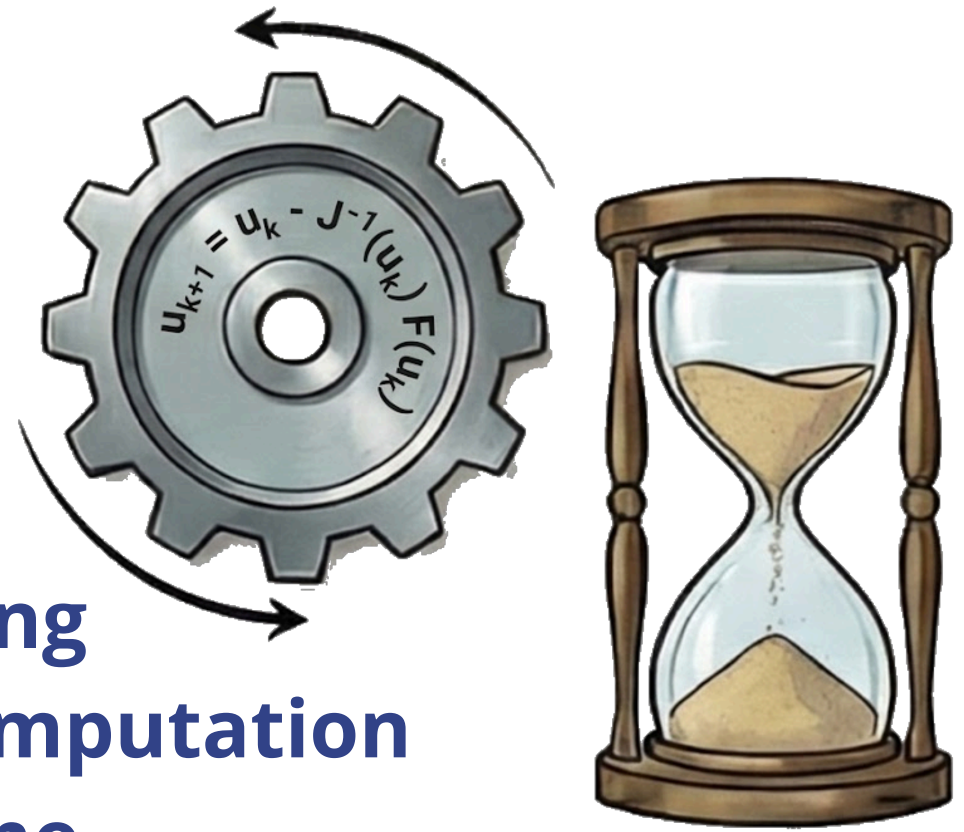
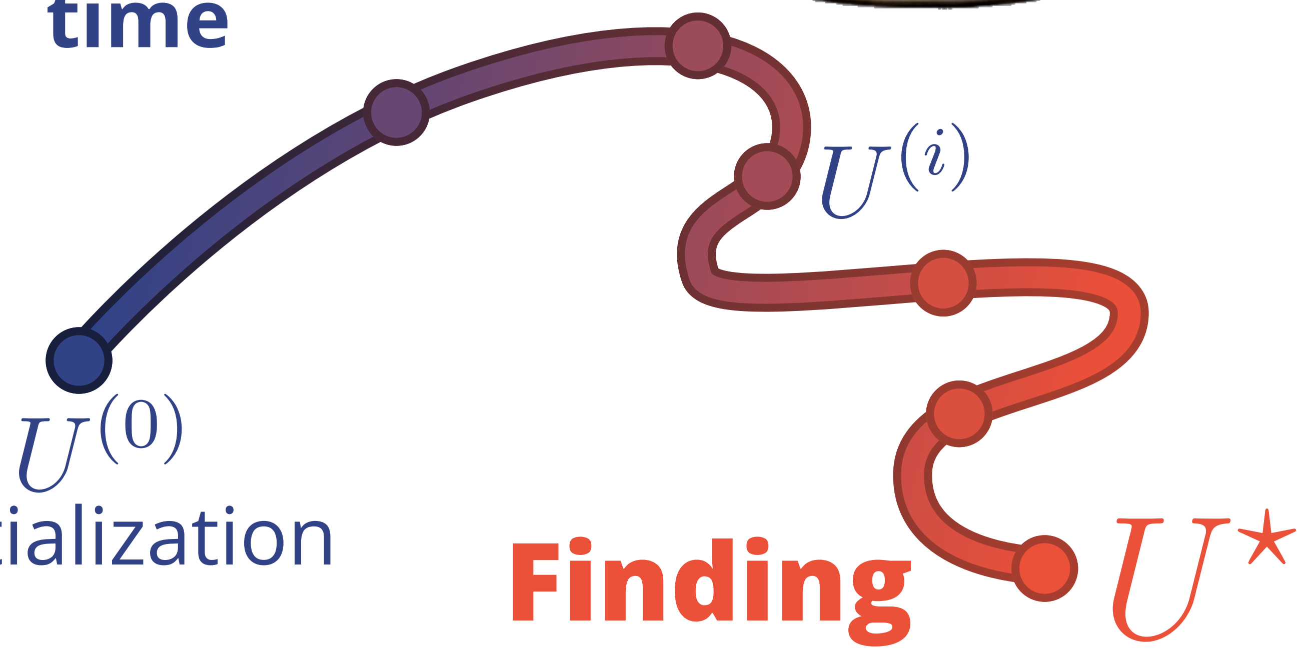
Traditional solvers



Long computation time



Finding $U^{\star}$ such that

$$F(U^{\star}, \lambda) = 0$$

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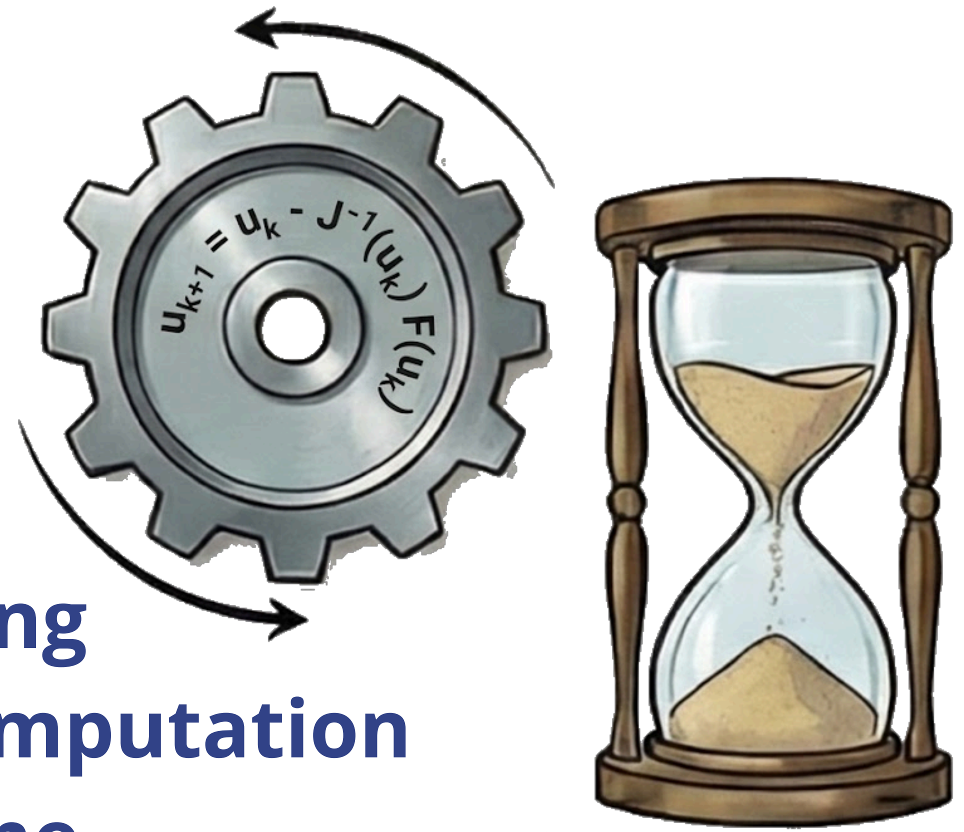

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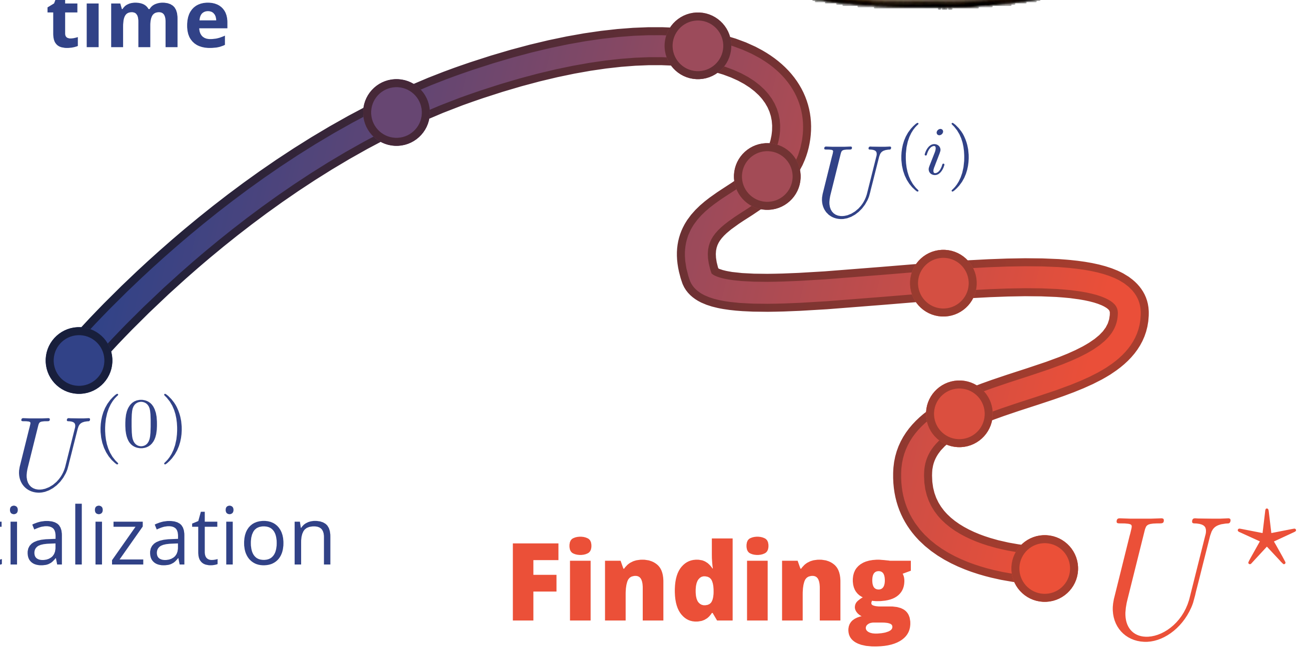
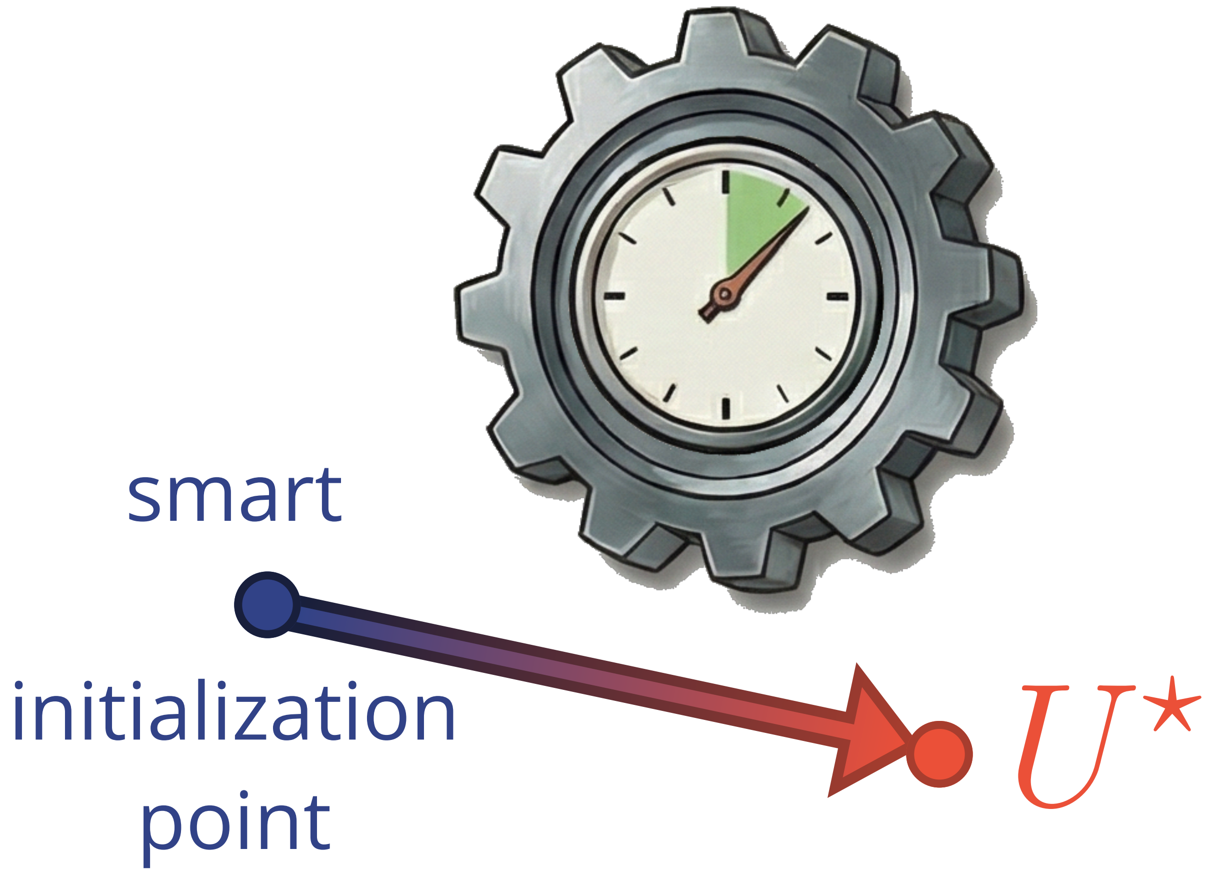


Finding $U^\star$ **such that** $F(U^\star, \lambda) = 0$

Initialization strategy



Applied methods

- **Nearest solution in train set**
- **Proper Orthogonal Decomposition**
- **Neural Network**
- **DeepONet**

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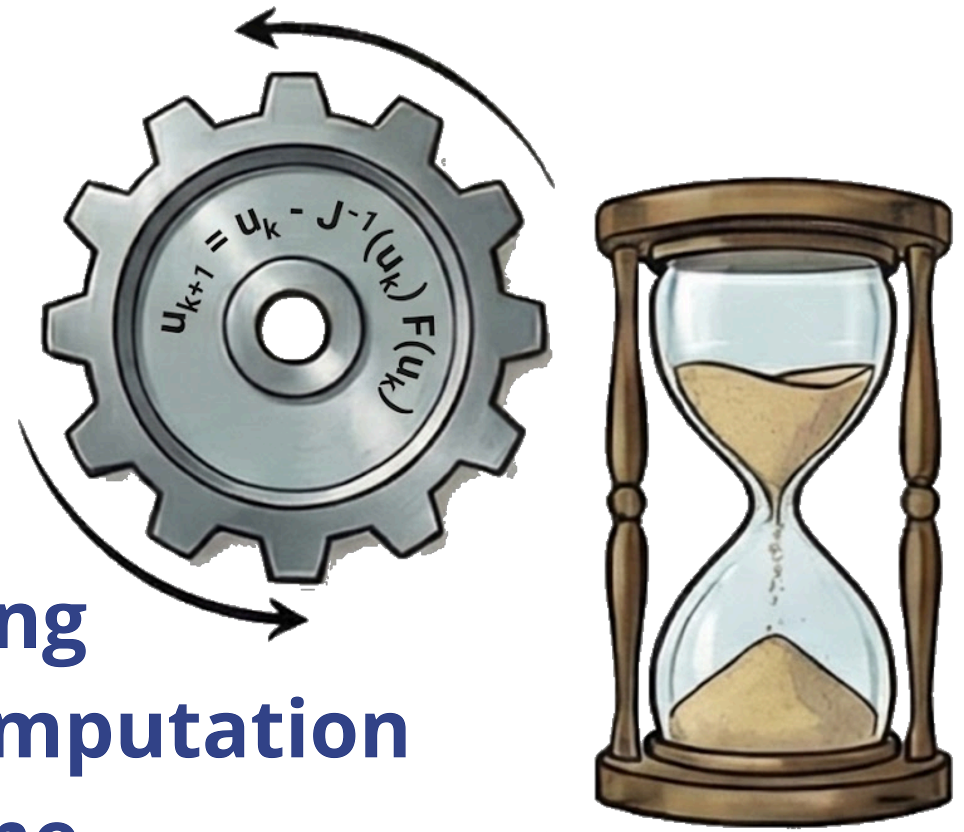

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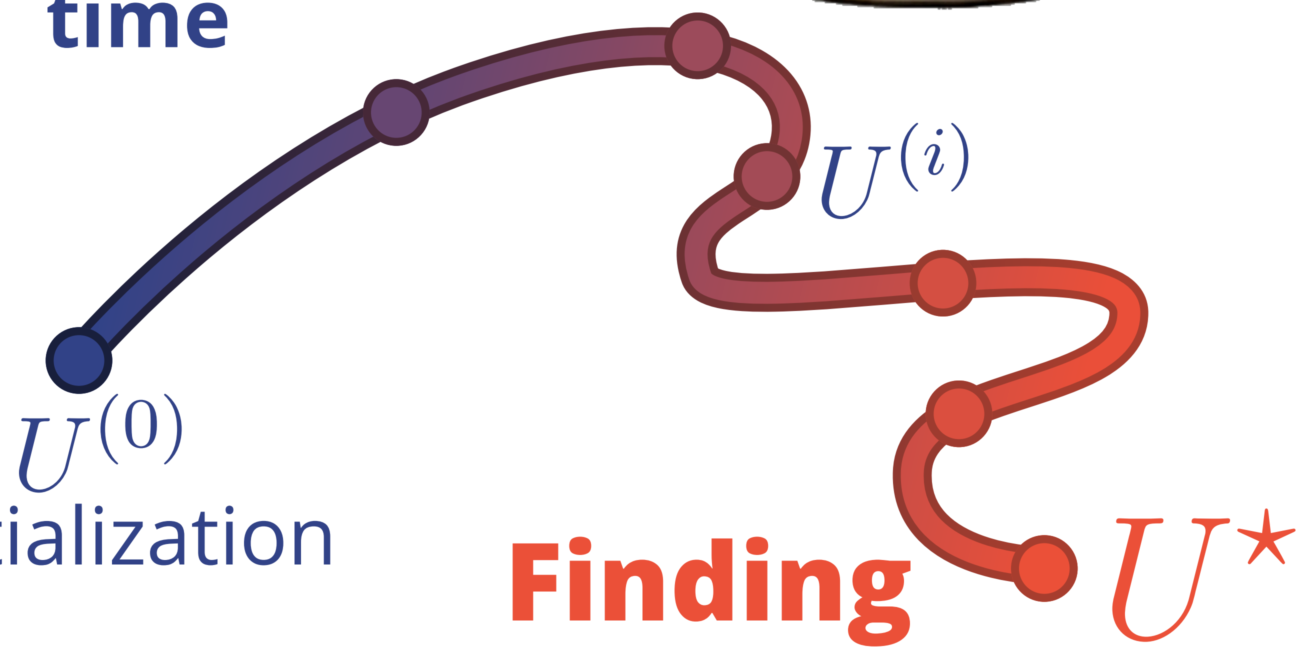

initialization

Finding $U^\star$ **such that**

$$F(U^\star, \lambda) = 0$$

Initialization strategy

smart initialization point

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Use cases:

1D Nonlinear Poisson equation

$$\begin{cases} -\frac{\partial}{\partial x}\left[q(u)\frac{\partial}{\partial x}u\right] = g(x,\lambda) & \text{in } \Omega \\ u = u_D & \text{on } \partial\Omega \end{cases}.$$

Calendering process

Rubber process manufacturing : compress material between two counter-rotating rolls.

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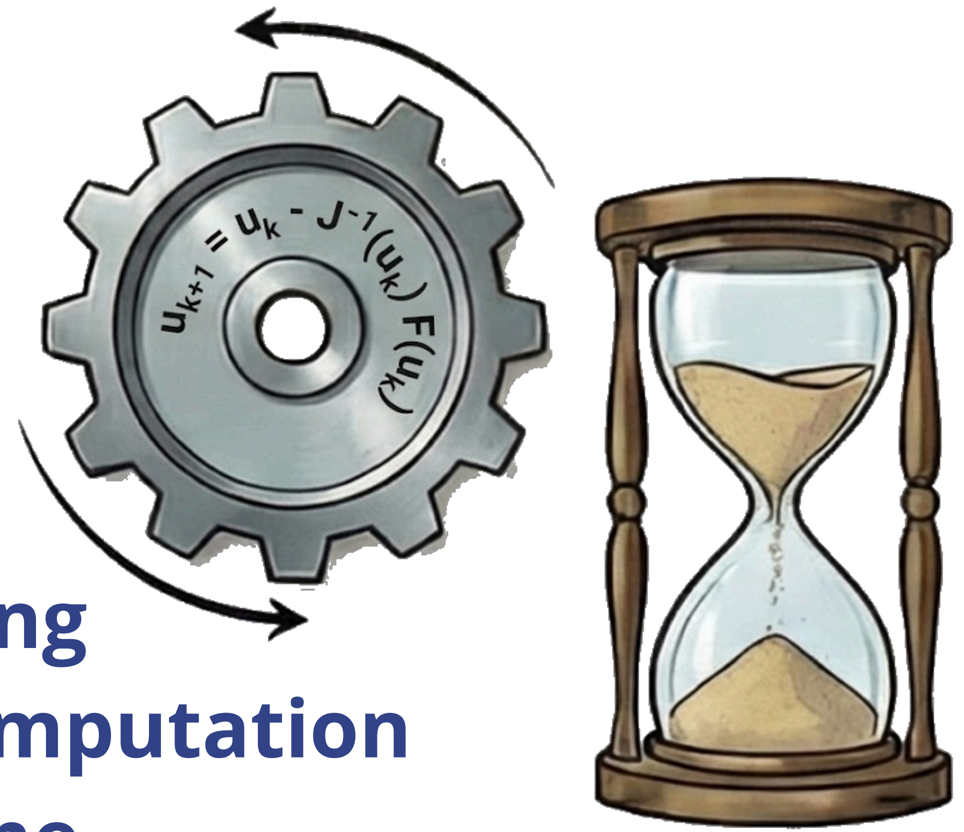

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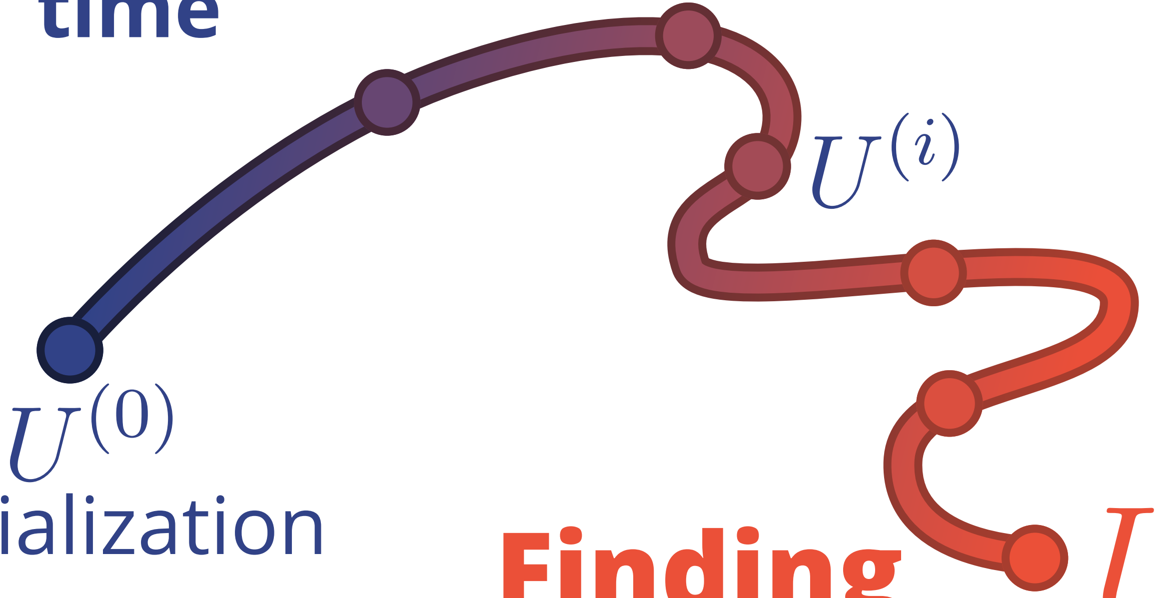
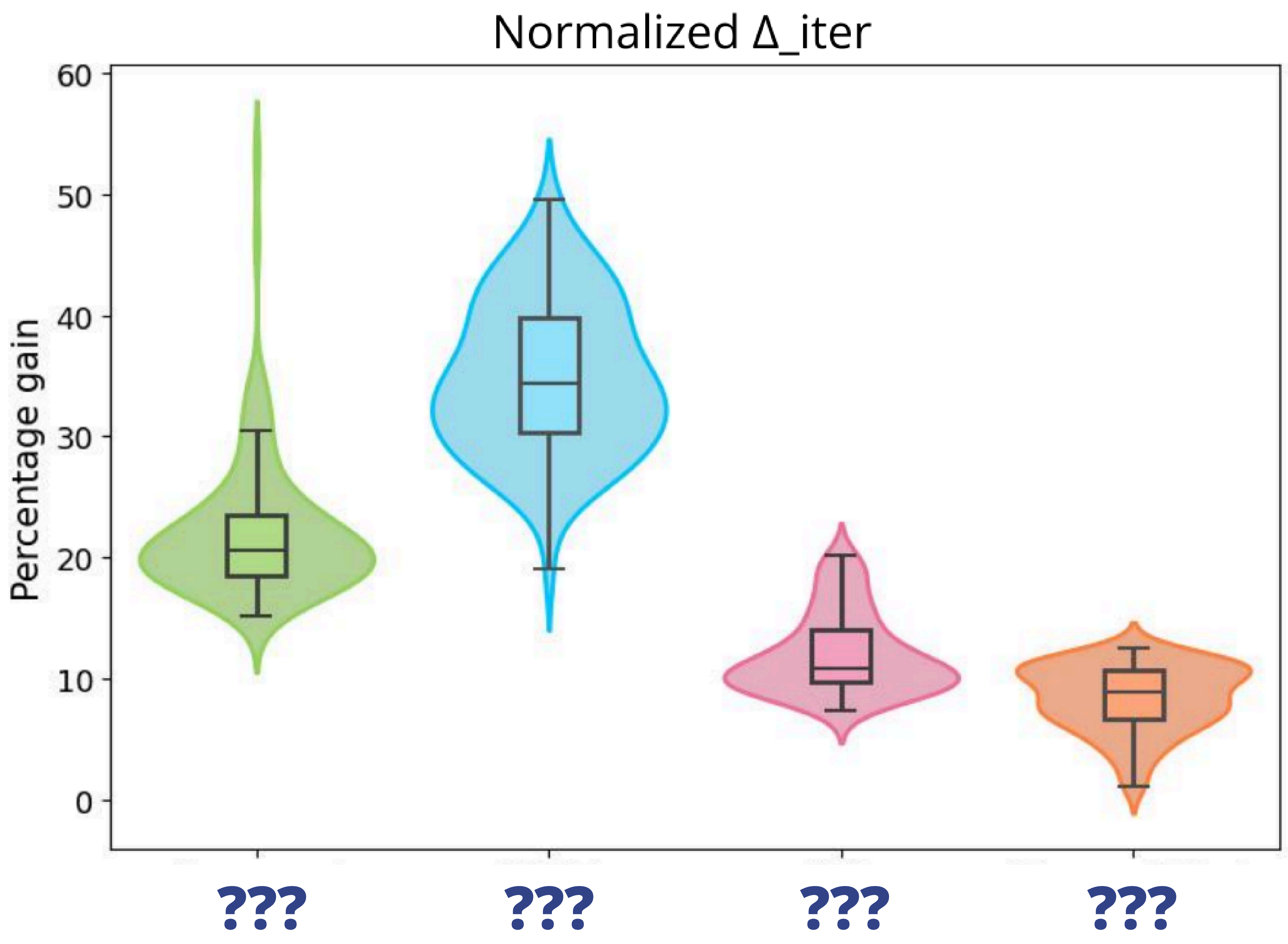


$U^{(0)}$ initialization

Finding $U^\star$ **such that**

$$F(U^\star, \lambda) = 0$$



Initialization strategy

smart initialization point $\longrightarrow U^\star$

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